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*Lab Assignment 01*  
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*FA25-BCS-031*  
*Data Structure*  
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Program #01

Code:

```
#include <iostream>

using namespace std;

int arr[100];

int arrSize = 0;

// 1. Insert value at end
void insertEnd(int value)
{
    if (arrSize < 100)
    {
        arr[arrSize] = value;
        arrSize++;
        cout << "Value inserted at end.\n";
    }
    else
    {
        cout << "List is full.\n";
    }
}

// 2. Insert value at start
void insertStart(int value)
{
    if (arrSize < 100)
    {
        for (int i = arrSize; i > 0; i--)
```

```
{  
    arr[i] = arr[i - 1];  
}  
  
arr[0] = value;  
arrSize++;  
  
cout << "Value inserted at start.\n";  
}  
else  
{  
    cout << "List is full.\n";  
}  
}
```

*// 3. Insert value after specific value*

*void insertAfter(int key, int value)*

```
{  
    if (arrSize >= 100)  
    {  
        cout << "List is full.\n";  
        return;  
    }  
  
    for (int i = 0; i < arrSize; i++)  
    {  
        if (arr[i] == key)  
        {  
            for (int j = arrSize; j > i + 1; j--)
```

```

    {
        arr[j] = arr[j - 1];
    }

    arr[i + 1] = value;
    arrSize++;

    cout << "Value inserted after " << key << ".\n";
    return;
}

cout << "Specific value not found.\n";
}

// 4. Insert value before specific value
void insertBefore(int key, int value)
{
    if (arrSize >= 100)
    {
        cout << "List is full.\n";
        return;
    }

    for (int i = 0; i < arrSize; i++)
    {
        if (arr[i] == key)
        {
            for (int j = arrSize; j > i; j--)

```

```
{  
    arr[j] = arr[j - 1];  
}  
  
arr[i] = value;  
arrSize++;  
  
cout << "Value inserted before " << key << ".\n";  
return;  
}  
}  
  
cout << "Specific value not found.\n";  
}  
  
// 5. Display array list  
void display()  
{  
    if (arrSize == 0)  
    {  
        cout << "List is empty.\n";  
        return;  
    }  
  
    cout << "Array List: ";  
  
    for (int i = 0; i < arrSize; i++)  
    {  
        cout << arr[i] << " ";  
    }
```

```
}
```

```
cout << endl;
```

```
}
```

```
// 6. Delete value from end
```

```
void deleteEnd()
```

```
{
```

```
    if (arrSize == 0)
```

```
    {
```

```
        cout << "List is empty.\n";
```

```
        return;
```

```
    }
```

```
    arrSize--;
```

```
    cout << "Value deleted from end.\n";
```

```
}
```

```
// 7. Delete value from start
```

```
void deleteStart()
```

```
{
```

```
    if (arrSize == 0)
```

```
    {
```

```
        cout << "List is empty.\n";
```

```
        return;
```

```
    }
```

```
    for (int i = 0; i < arrSize - 1; i++)
```

```
{  
    arr[i] = arr[i + 1];  
}  
  
arrSize--;  
  
cout << "Value deleted from start.\n";  
}
```

*// 8. Delete specific value*

```
void deleteSpecific(int value)  
{  
    if (arrSize == 0)  
    {  
        cout << "List is empty.\n";  
        return;  
    }  
  
    for (int i = 0; i < arrSize; i++)  
    {  
        if (arr[i] == value)  
        {  
            for (int j = i; j < arrSize - 1; j++)  
            {  
                arr[j] = arr[j + 1];  
            }  
  
            arrSize--;  
        }  
    }  
}
```

```

        cout << "Value deleted successfully.\n";
        return;
    }
}

```

```

    cout << "Value not found.\n";
}

```

```

// Main function

```

```

int main()
{
    int choice;
    int value;
    int key;

    do
    {
        cout << "\n===== ARRAY LIST =====\n";
        cout << "1. Insert value at end\n";
        cout << "2. Insert value at start\n";
        cout << "3. Insert value after specific value\n";
        cout << "4. Insert value before specific value\n";
        cout << "5. Display array list\n";
        cout << "6. Delete value from end\n";
        cout << "7. Delete value from start\n";
        cout << "8. Delete specific value\n";
        cout << "9. Exit\n";
        cout << "===== \n";
    }
}

```

```
cout << "Enter your choice: ";
```

```
cin >> choice;
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        cout << "Enter value: ";
```

```
        cin >> value;
```

```
        insertEnd(value);
```

```
        break;
```

```
    case 2:
```

```
        cout << "Enter value: ";
```

```
        cin >> value;
```

```
        insertStart(value);
```

```
        break;
```

```
    case 3:
```

```
        cout << "Enter specific value: ";
```

```
        cin >> key;
```

```
        cout << "Enter value to insert: ";
```

```
        cin >> value;
```

```
        insertAfter(key, value);
```

```
        break;
```

```
    case 4:
```

```
        cout << "Enter specific value: ";
```

```
cin >> key;
```

```
cout << "Enter value to insert: ";
```

```
cin >> value;
```

```
insertBefore(key, value);
```

```
break;
```

```
case 5:
```

```
display();
```

```
break;
```

```
case 6:
```

```
deleteEnd();
```

```
break;
```

```
case 7:
```

```
deleteStart();
```

```
break;
```

```
case 8:
```

```
cout << "Enter value to delete: ";
```

```
cin >> value;
```

```
deleteSpecific(value);
```

```
break;
```

```
case 9:
```

```
cout << "Program ended.\n";
```

```
break;
```

```
default:
```

```
cout << "Invalid choice.\n";
```

```
}
```

```
} while (choice != 9);
```

```
return 0;
```

```
}
```

**OUTPUT:**

```
===== ARRAY LIST =====
1. Insert value at end
2. Insert value at start
3. Insert value after specific value
4. Insert value before specific value
5. Display array list
6. Delete value from end
7. Delete value from start
8. Delete specific value
9. Exit
=====
Enter your choice: 1
Enter value: 34
Value inserted at end.

===== ARRAY LIST =====
1. Insert value at end
2. Insert value at start
3. Insert value after specific value
4. Insert value before specific value
5. Display array list
6. Delete value from end
7. Delete value from start
8. Delete specific value
9. Exit
=====
Enter your choice: 2
Enter value: 23
Value inserted at start.

===== ARRAY LIST =====
1. Insert value at end
2. Insert value at start
3. Insert value after specific value
4. Insert value before specific value
5. Display array list
6. Delete value from end
7. Delete value from start
8. Delete specific value
9. Exit
```

## Program #02

*Code:*

```
#include <iostream>

using namespace std;

int main()
{
    int start, stop;

    int sum = 0;

    cout << "Enter starting value: ";
    cin >> start;

    cout << "Enter stopping value: ";
    cin >> stop;

    for(int x = start; x <= stop; x++)
    {
        sum = sum + (x * x);
    }

    cout << "Sum of squares = " << sum;

    return 0;}
```

 C:\Users\hp\OneDrive\Desktop\program02.exe

```
Enter starting value: 6
Enter stopping value: 90
Sum of squares = 247010
-----
Process exited after 18.11 seconds with return value 0
Press any key to continue . . .
```

**Program #03****Code:**

```
#include <iostream>

using namespace std;

int main()
{
    int arr[100];
    int size;
    int value;

    cout << "Enter size of Array List: ";
    cin >> size;

    cout << "Enter elements:\n";

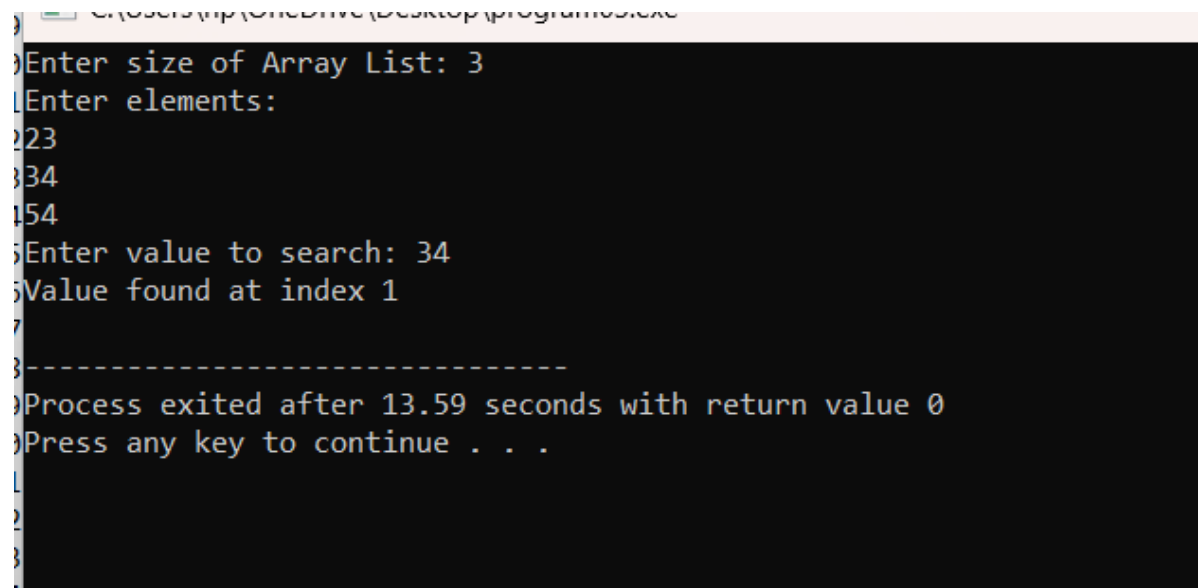
    for(int i = 0; i < size; i++)
    {
        cin >> arr[i];
    }

    cout << "Enter value to search: ";
    cin >> value;

    int i = 0;
    bool found = false;

    while(i < size)
```

```
{  
    if(arr[i] == value)  
    {  
        cout << "Value found at index " << i << endl;  
        found = true;  
        break;  
    }  
  
    i++;  
}  
  
if(found == false)  
{  
    cout << "Value not found.";  
}  
  
return 0;  
}
```



The screenshot shows a terminal window with the following output:

```
0) Enter size of Array List: 3  
1) Enter elements:  
2) 23  
3) 34  
4) 54  
5) Enter value to search: 34  
5) Value found at index 1  
7)  
8) -----  
9) Process exited after 13.59 seconds with return value 0  
9) Press any key to continue . . .  
1)  
2)  
3)
```